



An Open-Source, Low-Cost Automated Optical Inspection System for PCB Defect Detection Using YOLOv8 and Custom Robotics

Abstract— Automated Optical Inspection (AOI) systems are critical for Printed Circuit Board (PCB) quality control, yet commercial solutions remain cost-prohibitive for small-scale manufacturing and educational laboratories. This paper presents the design, implementation, and evaluation of an open-source, ultra-low-cost (approx. 6000 INR) AOI system. The architecture integrates a custom 3D-printed single-axis gantry, an ESP32 microcontroller, and a high-resolution CMOS sensor (Samsung ISOCELL HP2). Defect detection is powered by a YOLOv8-Medium deep learning model optimized on an NVIDIA RTX 3050 Ti GPU, achieving a real-time inference speed of 13.0 ms. The system was trained on a custom dataset of manually altered Arduino Uno boards, simulating industrial lighting environments. Empirical results demonstrate exceptional model performance, achieving a mean Average Precision (mAP@0.5) of 0.995 across 8 distinct missing-component classes with zero latency in live sweep operations.

Keywords— Automated Optical Inspection, PCB Defect Detection, YOLOv8, Computer Vision, Gantry Robotics, Open-Source Hardware.

I. INTRODUCTION

The proliferation of open-source electronics and small-scale PCB manufacturing has democratized hardware engineering. However, quality assurance remains a significant bottleneck. Missing surface-mount devices (SMDs), misaligned integrated circuits (ICs), and omitted headers often lead to catastrophic circuit failures. Traditional Automated Optical Inspection (AOI) systems rely on rigid machine vision algorithms and expensive proprietary hardware, often exceeding \$10,000 USD, rendering them inaccessible to DIY enthusiasts, educational labs, and micro-factories.

Recent advancements in Convolutional Neural Networks (CNNs), specifically the You Only Look Once (YOLO) architecture, have enabled real-time object detection using standard optical hardware. This research proposes a fully reproducible, low-cost AOI system utilizing a YOLOv8m model paired with a custom robotic gantry. By leveraging ubiquitous consumer electronics (a smartphone camera network node) and 3D-printed mechanics, the system simulates industrial-grade quality control for approximately 6000 INR (~\$72 USD).

II. SYSTEM ARCHITECTURE

The proposed inspection system is divided into three distinct subsystems: Mechanical, Electronic/Control, and Computational/Software.

A. Mechanical Gantry Design

To facilitate scanning across the PCB, a custom single-axis linear gantry was fabricated using 3D-printed structural components. The gantry provides a clear optical travel length of 220 mm. The camera carriage is driven by a timing belt connected to an MG996R continuous rotation servo motor, selected for its high torque capabilities within a low-cost budget constraint.

B. Electronic Control Node

The physical actuation is managed by an ESP32 microcontroller. To safely isolate logic power from motor draw, an I2C-driven PCA9685 PWM servo driver is utilized, powered independently by a

dedicated 5V, 3A adapter. Hardware limit switches are placed at both ends of the 220 mm axis, providing instant closed-loop safety interrupts to the ESP32.

C. Optical and Computational Pipeline

The optical sensor utilized is a 200MP Samsung ISOCELL HP2 (housed within a Galaxy S23 Ultra), functioning as an IP streaming node over a local wireless network. The central processing unit is a laptop equipped with an NVIDIA RTX 3050 Ti GPU. A custom graphical user interface (GUI) developed in PyQt6 bridges the computational and control nodes. A key feature of this GUI is the "Auto Mode," which commands the ESP32 to continuously sweep the camera left and right across the 220 mm axis, creating a zero-lag, continuous detection loop.

149	2584.66	0.6726	0.37808	1.15211	0.97655	1	0.995	0.74595	0.42188	0.21036	0.63183
150	2601.79	0.70112	0.3857	1.19365	0.98109	1	0.995	0.74863	0.41992	0.20833	0.63063

III. DATASET PREPARATION AND METHODOLOGY

The most significant challenge in deep learning-based AOI is acquiring a dataset that accurately reflects the physical conditions of the deployment environment, particularly regarding illumination and component scale.

A. Data Collection

To simulate an industrial inspection line, multiple defective Arduino Uno boards were created by manually de-soldering specific components. Images were captured against a high-contrast white paper background under an artificially diffused lighting setup to eliminate harsh shadows and glare. A total of 178 raw baseline images were collected, categorized into 8 defect classes corresponding to missing components: Analog Pins, ATmega IC, Capacitors, Crystal Oscillator, Digital Pins, EPS, USB Port, and Voltage Regulator.

B. Augmentation Strategy

Given the limited size of the baseline dataset, morphological and photometric augmentations were applied to ensure model robustness. Augmentations included random rotation ($\pm 15^\circ$), brightness variation ($\pm 20\%$), and Gaussian blur (up to 1.5px) to simulate potential gantry vibrations. This expanded the dataset to over 500 images, partitioned into 70% training, 20% validation, and 10% testing subsets.

C. YOLOv8 Training Configuration

The YOLOv8 Medium (YOLOv8m) architecture was selected to balance deep feature extraction with real-time inference speeds. The model was trained over 150 epochs with a batch size of 16 and an image resolution of 640x640.

IV. EXPERIMENTAL RESULTS

The system was evaluated on both historical validation data and live physical test runs. The model exhibited exceptional convergence, with bounding box and classification loss stabilizing significantly by Epoch 120.

A. Precision and Recall Metrics

149	2584.66	0.6726	0.37808	1.15211	0.97655	1	0.995	0.74595	0.42188	0.21036	0.63183
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150 2601.79 0.70112 0.3857 1.19365 0.98109 1 0.995 0.74863 0.41992 0.20833 0.63063

As detailed in Table I, the model achieved a near-perfect Mean Average Precision. Notably, the system detected macro-components (e.g., ATmega chip) and micro-components (e.g., crystal oscillator) with equal accuracy, proving the effectiveness of the chosen 640x640 resolution padding strategy.

TABLE I: YOLOv8 EVALUATION METRICS (EPOCH 150)

Metric	Value
mAP@0.5 (All Classes)	0.995
mAP@0.5-0.95 (All Classes)	0.748
Precision (B)	0.981
Recall (B)	1.000

B. Confusion Matrix Analysis

Analysis of the validation confusion matrix confirmed zero instances of background false positives (ghosting) and zero misclassifications between defect classes.

TABLE II: VALIDATION INSTANCE ACCURACY

Defect Class	Instances	True Positives	False Positives
missing_analog_pins	5	5	0
missing_atmega	11	11	0
missing_capacitors	13	13	0
missing_crystal_os	5	5	0
missing_digital_pins	6	6	0
missing_eps	7	7	0
missing_usb	9	9	0
missing_voltage_reg	3	3	0

C. Live System Performance

During live execution in "Auto Mode," the RTX 3050 Ti processed incoming IP camera frames with an inference latency of **13.0 ms per image** (equivalent to ~76 frames per second). This computational headroom resulted in a visually seamless operator experience with zero perceptible lag between physical gantry movement and UI bounding box updates.

V. CONCLUSION AND FUTURE WORK

This paper demonstrates that industrial-grade Automated Optical Inspection is achievable using open-source deep learning frameworks and low-cost 3D-printed robotics. For an approximate build cost of 6000 INR, the proposed YOLOv8-based system successfully automates PCB defect detection with 99.5% average precision and real-time execution speeds.

Future work will focus on scaling the mechanical architecture to support high-throughput environments. We propose the development of an intelligent dual-belt conveyor system to replace the single-axis gantry. This continuous-feed system will allow multiple cameras to inspect different boards or different sides of the same PCB simultaneously, further bridging the gap between DIY hardware and commercial manufacturing standards.

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